# Ivan Zolin

Research Engineer

Multimodal Learning · Efficient ML · Multi-agent Systems · LLMs

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## Summary

Research Engineer building machine learning systems across LLMs/NLP, computer vision, and multimodal learning, with a focus on efficient models, multi-agent systems, and adaptive architectures.

4+ years of research and engineering experience, taking ML methods from experimentation and model training to optimization and production deployment, with international conference recognition and awards from research and innovation programs.

**Skills:**

- Python, C/C++, SQL, PyTorch, JAX, TensorFlow, ONNX, LangChain, LangGraph, CUDA, Docker
- Machine Learning, Deep Learning, Multi-Modal Learning, Algorithms and Data Structures

**Research Interests:** Multimodal Learning, Efficient ML, Multi-agent Systems, LLMs, RAG, Knowledge Graph, Semantic Alignment, Multimodal Representation Learning

## Experience

**NLP R&D Engineer**, Digital Public Health Technologies, ITMO University — Aug 2025 – Present

- Designed and built the orchestrated MAS-EHR pipeline for OCR, de-identification, structuring, and ontology extraction, running as chunked asynchronous stages with retries and per-stage observability; led the design and development of its HITL and interpretation modules, defining architecture, workflows, and validation logic while coordinating implementation across the team.
- Reached 0.89 precision at 0.86 recall de-identifying unstructured medical text, and about 95% text similarity with 75% exact match recognising scanned medical documents.

**AI Researcher (CV & NLP)**, Laboratory of Biomedical Imaging and Data Analysis, SPbPU — Apr 2024 – Apr 2026

- Developed TriDeFusion-v2, a hybrid spatial-frequency restoration framework for volumetric fluorescence microscopy pairing a distilled attention CNN with locally adaptive wavelet refinement, raising PSNR to 29.2 dB from 22.6 dB for TriDeFusion-v1 on high-noise synthetic tubular volumes at up to 3x faster inference; published.
- Developed NeuroRAG, a multi-agent LLM and RAG system for neurobiological question answering over a curated vector store, PubMed, arXiv and NCBI, whose answers 13 domain specialists preferred to GPT-5 and Claude 4.5 Sonnet in 72.6% of 135 decisive blind comparisons; published.

**Junior CV Researcher**, Laboratory of Molecular Neurodegeneration, SPbPU — Apr 2023 – Apr 2024

- Developed TriDeFusion, a 3D fluorescence-microscopy denoising algorithm combining Noise2Noise and Non-local Means, cutting RMSE by up to 54% and raising PSNR by 20% over the N2N-UNet baseline on synthetic data; presented at IEEE SIBIRCON 2024.
- Developed FRT, a web-based system for processing 3D microscopy images using machine learning and deep learning methods.

## Education

**M.S.**, ITMO University — Aug 2024 – Jun 2026
Deep Learning and Generative Artificial Intelligence • GPA: 4.90/5.00
*Thesis:* "Multi-agent Medical Records Data Processing System (EHR)"

**B.S.**, Peter the Great St. Petersburg Polytechnic University — Sep 2020 – Jun 2024
Applied Mathematics and Computer Science • GPA: 4.53/5.00
*Thesis:* "Online AI-based deconvolution and denoising service for confocal microscopy"

## Projects

**MAS-EHR** [MAS-EHR Presentation] — Aug 2025 – Present
LangGraph, MAS, Knowledge Graph, QLoRA, Qwen, Gemma, OCR, NER, PII, HITL

- MAS-EHR: a multi-agent system turning heterogeneous medical documents into a structured patient record — a layout-aware OCR agent on Qwen3.5:9B fine-tuned with QLoRA over an in-house corpus of form templates, real clinical records and handwriting; a de-identification agent combining regular expressions, NER, an LLM and configured relations between entity pairs; and a structuring agent normalising symptoms, diagnoses, drugs, labs and procedures against SNOMED CT, ICD-10 and LOINC.

- Ontology agent representing each patient as a dynamic knowledge graph: extracted entities are clustered and aligned by embedding similarity, new entity and relation types are discovered as records arrive, and the graph is served through Graph RAG so retrieval follows clinical relations rather than text similarity alone.

**NeuroRAG** [GitHub] Apr 2024 – Present
RAG, MAS, LangGraph, LangChain, RRF, BM25, HyDE, RAPTOR, Chroma, Contextual Compression, Tavily, Biopython, Pydantic, OpenRouter, Docker, FastAPI, Streamlit

- NeuroRAG: a multi-agent LLM and RAG system for neurobiological question answering, built as a LangGraph state machine that routes each query across a curated vector store, PubMed, arXiv and NCBI, expands it four ways, then filters by Reciprocal Rank Fusion, BM25 and LLM reranking before answering with three LLMs in parallel under grounding and relevance checks.
- Highest accuracy on seven of nine biomedical MMLU subsets (99.0% on Medical Genetics), 23–53 percentage points above BioMistral-7B on every subset, and ahead of a FlashRAG Naive RAG baseline sharing its generator on all six text metrics (BERTScore 0.425 against 0.206).

**TriDeFusion** [GitHub] Apr 2023 – Present
PyTorch, CUDA, CuPy, cuML, Noise2Noise, Knowledge Distillation, Flash Attention, CBAM, ECA, Attention Gates, PyWavelets, Optuna, LPIPS

- TriDeFusion (v1): 3D fluorescence-microscopy denoising algorithm combining Noise2Noise and Non-local Means; on real acquisitions it cut mean noise intensity by over 50% and noise variance by 33%.
- TriDeFusion-v2: RAUDen, a residual attention U-Net distilled into a compact student by Noise2Noise knowledge distillation and followed by a GPU wavelet stage whose thresholds adapt to local noise statistics, both configured by Bayesian optimisation; the most favourable quality-efficiency trade-off among classical, CNN, transformer and generative baselines across synthetic tubular volumes, confocal dendrite stacks and in vivo miniscope calcium recordings.

**FRT** [GitHub] Apr 2023 – Present
FastAPI, Celery, Redis, S3, React, Docker

- Web service for 3D microscopy image processing: a FastAPI backend serving the ML and DL models, Celery with Redis for long-running jobs, S3 for volume storage, and a React interface.

## Awards and Honors

- International AI & Big Data Award "Gravity 2026": Winner, Youth Track, for MAS-EHR. [Certificate]
- International AI & Big Data Award "Gravity 2025": Winner and finalist, for "Decoding the Brain" (Breakthrough R&D) and "NeuroRAG" (Youth Track). [Certificate] [Certificate]
- Pitch the Future: Med AI: Recipient of the Selectel grant, with the MAS-EHR project. [Event page] 2026

## Talks

- ODS Data Fest 2026: Speaker in the "Medical Data Horror Stories" track with the talk "MAS-EHR: A Multi-Agent System for Clinical Data Integration and Harmonization". [Slides]
- International ITMO Conference "Young Professionals": Participant. [Certificate] 2025

## Publications

**TriDeFusion-v2: Distilled Hybrid Residual Attention Denoising and Wavelet Filtering** 2026 [IEEE Xplore]
**Ivan Zolin**, Anastasiya Rakovskaya, Evgenii Fedorov, Sergey Belyaev, Vyacheslav Chukanov, Ekaterina Pchitskaya
IEEE Access

**NeuroRAG: A Multi-Agent Model for Neurobiological Question Answering** 2026 [MDPI]
Vladimir Skvortsov, **Ivan Zolin**, Vyacheslav Chukanov, Ekaterina Pchitskaya
Big Data and Cognitive Computing

**TriDeFusion: Enhanced denoising algorithm for 3D fluorescence microscopy images integrating modified Noise2Noise and Non-local means** 2024 [IEEE Xplore] [pdf]
**Ivan Zolin**, Vyacheslav Chukanov, Ekaterina Pchitskaya
2024 IEEE International Multi-Conference on Engineering, Computer and Information Sciences, SIBIRCON 2024

## Intellectual Property

- Detection and Anonymization of Personal Data in Unstructured Medical Texts [Registration]
- AI System Based on an Ensemble of Language Models with RAG Access to a Neuroscience Knowledge Base [Registration]
- Automatic Segmentation of Fluorescent Microspheres in Confocal Microscopy Images [Registration]
- Online AI-Based Deconvolution and Denoising Service for Confocal Fluorescence Microscopy Images [Registration]